

# Script of the Design Course

## *Design Course: RF- and Microwave Circuits and Systems*

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19.04.2026

Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient  
high-frequency electronics  
Freiburg University

## Contents

|          |                                                                      |           |
|----------|----------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Abstract</b>                                                      | <b>6</b>  |
| <b>2</b> | <b>Today's Topics: Real Lines</b>                                    | <b>10</b> |
| 2.1      | Educational Targets . . . . .                                        | 10        |
| 2.2      | Real Lines and Components . . . . .                                  | 11        |
| 2.3      | Task 2 (Block 1): LineCalc . . . . .                                 | 15        |
| 2.4      | Microstrip and Stubs: Methods of Moments: Task 3 (Block 1) . . . . . | 17        |

## Introduction

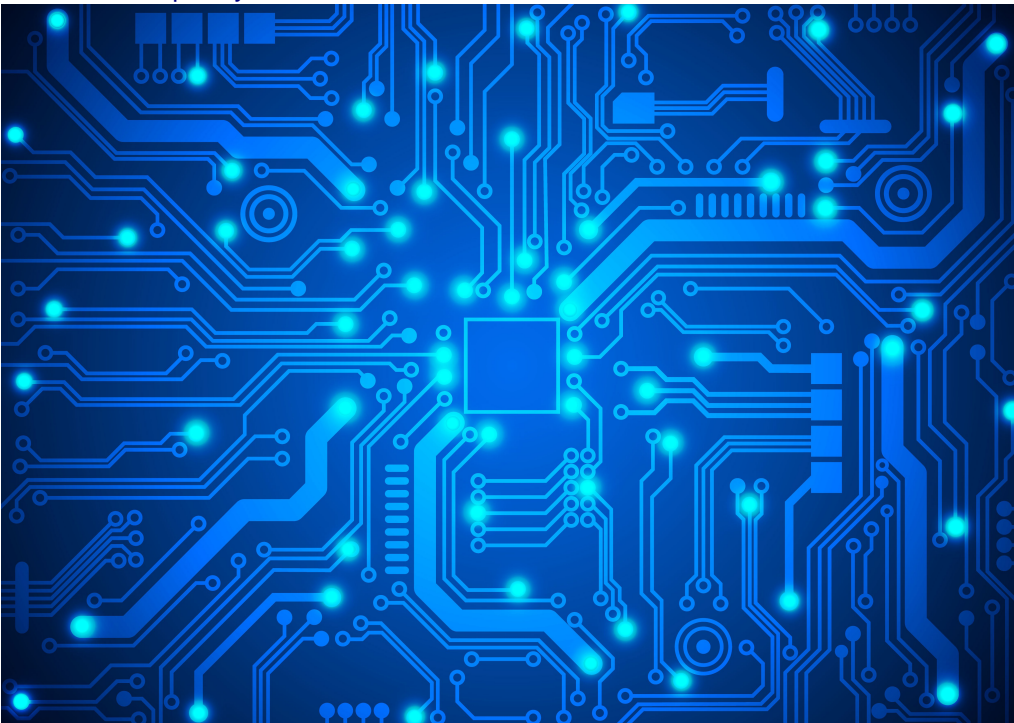


## Introduction



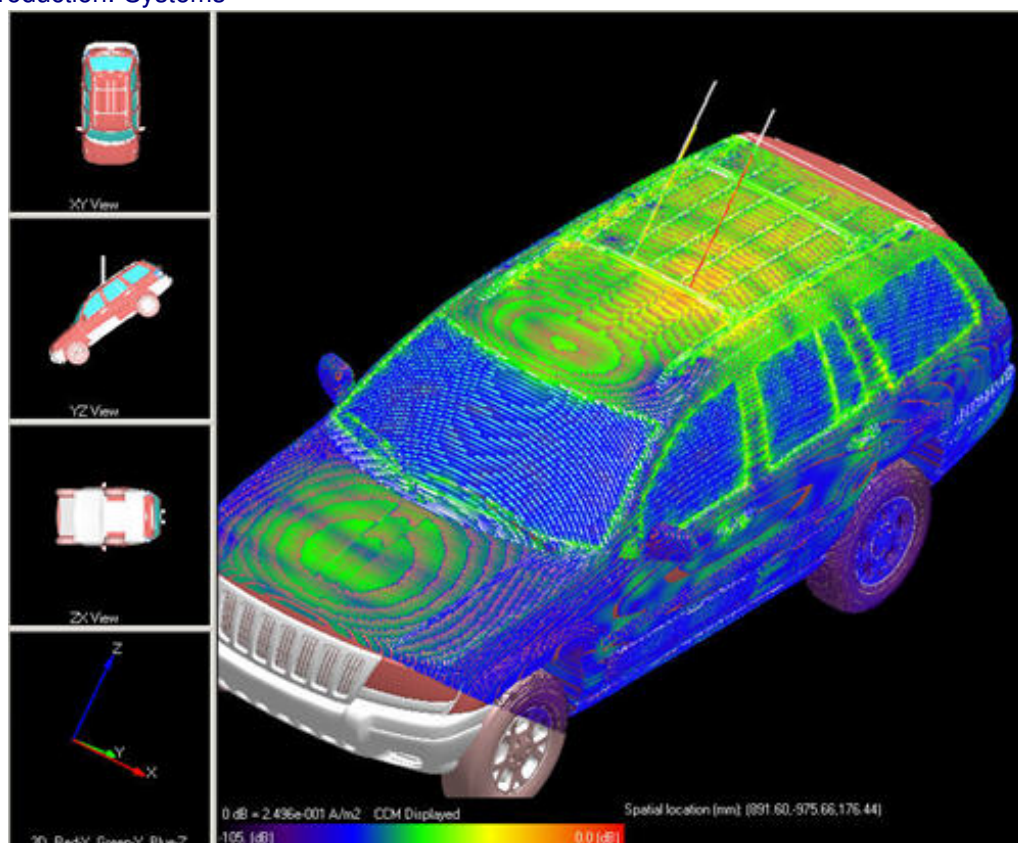
1.4

## Introduction: Complexity



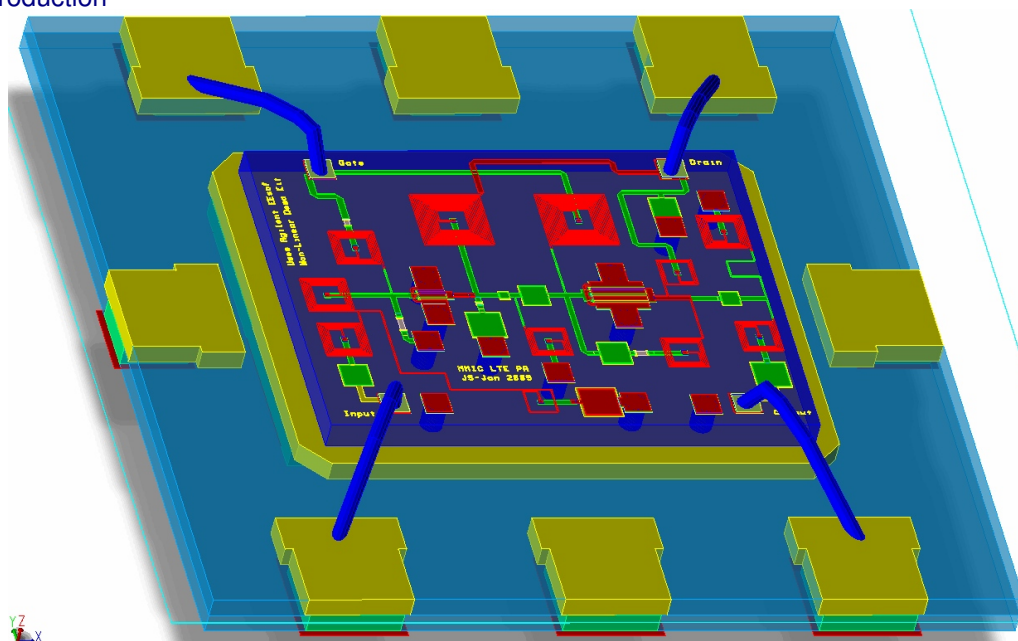
1.5

## Introduction: Systems



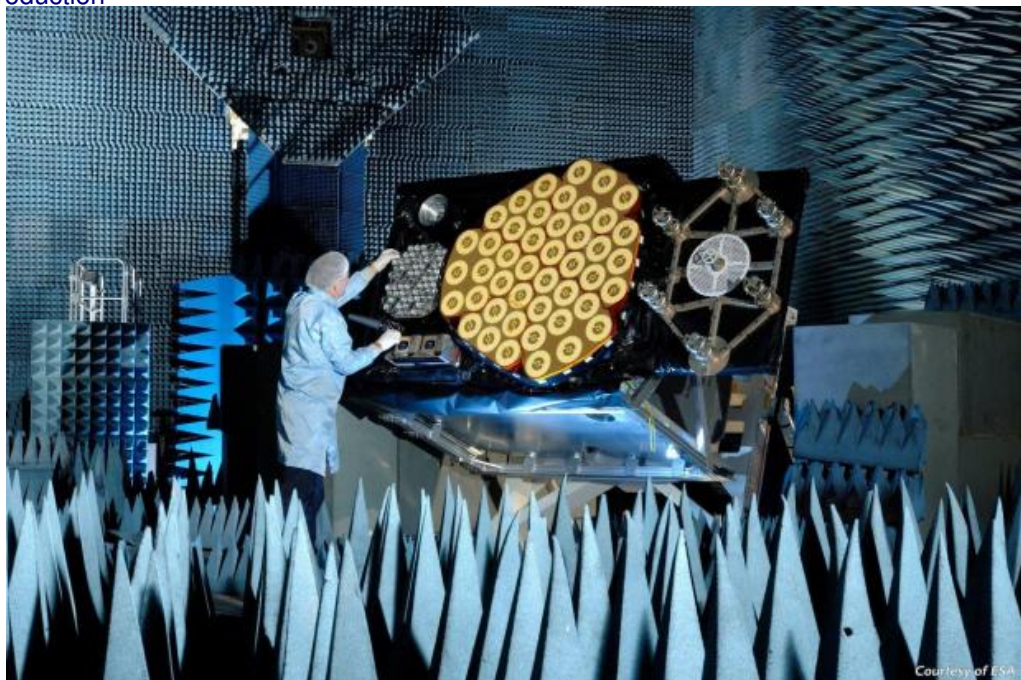
1.6

## Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



### Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 no lecture, pentecost break SR 00 014 (G.-Köhler-Allee 078)

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1.9

### Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

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1.10

## Date and Location

### Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 28.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 8.6.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 15.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 22.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 29.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 6.7.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 13.7.2026 :(issue block 5b ON ILIAS)
12. Mo., 20.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

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1.11

### Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

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1.12

## 1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

# Introduction

## Materials and Script

The material is provided at the beginning and during the lecture.

Email: [ruediger.quay@inatech.uni-freiburg.de](mailto:ruediger.quay@inatech.uni-freiburg.de)

- Foils: Intro materials from Keysight
- ADS Video Library on [www.youtube.com](http://www.youtube.com)
- Script (supporting materials)
- Exercises

1.13

## Videos

1. Watch ADS videos on youtube using ADS : [www.youtube.com](http://www.youtube.com) , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

1.14

## WWW Support 1

### ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

1.15

## WWW Support 2

This exercise is recorded, attendance is, however, required



1.16

## Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

1.17

## Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

1.18

## Install an ADS Version on a local computer

Last Updated: 05 March 2020

| ADS - Windows® Operating Systems                                                 |                                                                                        |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Product                                                                          | Windows                                                                                |
| ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0<br>64-bit Operating Systems | Windows 10 (64-bit)<br>Windows 7 Enterprise (64-bit)                                   |
| ADS 2019/<br>ADS 2019 Update 1.0<br>64-bit Operating Systems                     | Windows 10 (64-bit)<br>Windows 7 Enterprise (64-bit)                                   |
| ADS 2017/<br>ADS 2017 Update 1.0<br>64-bit Operating Systems                     | Windows 10 (64-bit)<br>Windows 8 Enterprise (64-bit)<br>Windows 7 Enterprise (64-bit)  |
| ADS 2016.01<br>64-bit Operating Systems                                          | Windows 10 (64-bit)*<br>Windows 8 Enterprise (64-bit)<br>Windows 7 Enterprise (64-bit) |
| ADS 2015.01<br>64-bit Operating Systems                                          | Windows 8 Enterprise (64-bit)<br>Windows 7 Enterprise (64-bit)                         |

**You can work on the aragorn but local has proven to be easier...**

**Organize your data in a good structure**

1.19

**Login to Application Server for the license**

**From Remote: Create RAS Access in Mylogin**

**Go into University net with VPN: check VPN Uni Freiburg**

**In case you use Windows to do that: use putty and install xming as windows X-server**

**Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`**

**Shells usable: `bash` and `tcsh`**

**Start: `ads` after using `ssh` to the server**

**Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`**

1.20

**Backup Server !**

**Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)**

1.21

## Recommendation

**Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0**

1.22

The following references may be useful to provide more detailed information.

## References

- List of useful ADS Videos
- Keysight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS\_Basics.pdf (ILIAS)
- Momentum Momentum\_Basics.pdf (ILIAS)

1.23

## Submission Exercise on ILIAS

- See exercise Block 1 Two tasks are combined in one block and submitted as one block
- 5 blocks Exercises with overall 10 topics/tasks
- First: starting this week... (two weeks to do...)
- 4 to be completed (with 50% correct) (three people per group)
- Competition: Final presentation (15 mins) of the group to the plenary based on block number 4b.

2.25

## 2 Todays Topics: Real Lines

### Introduction: Tasks and Exercises 1

1. Perform the exercise: Login remotely and Start ADS...
2. Work yourself through: ADS\_Basics.pdf (ILIAS)
3. Start Linecalc inside ADS and perform a small example of your choice
4. Perform the exercise: Start Momentum inside ADS
5. Work yourself through MOMENTUM in : Momentum\_Basics.pdf (ILIAS)

2.26

### 2.1 Educational Targets

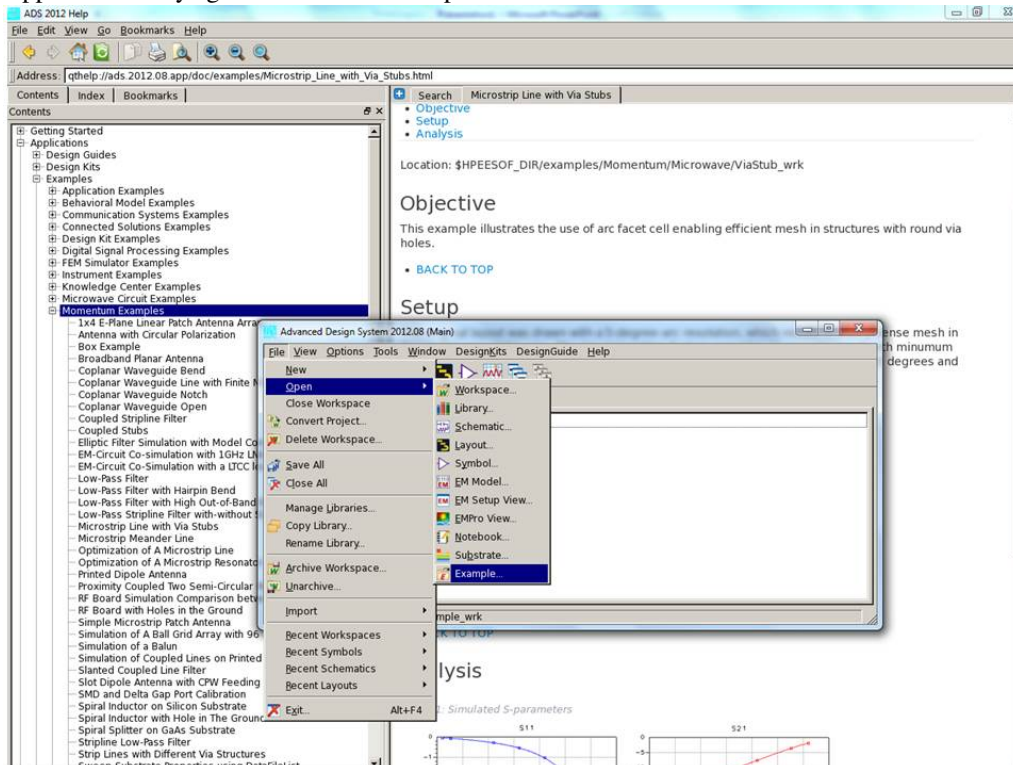
#### Educational Targets: Passive lines and components

- Part 1
- Block 1a: Understand and dimension a line with the simulator Linecalc inside ADS
- <https://www.youtube.com/watch?v=x8TSCnPSgDo>
- Block 1b: Perform a Momentum simulation with the method of moments for coupled microstrip lines to understand crosstalk and coupled lines !!!
- <https://www.youtube.com/watch?v=c0LXAkVKnDg>

2.27

### Where to find most of the files for ADS

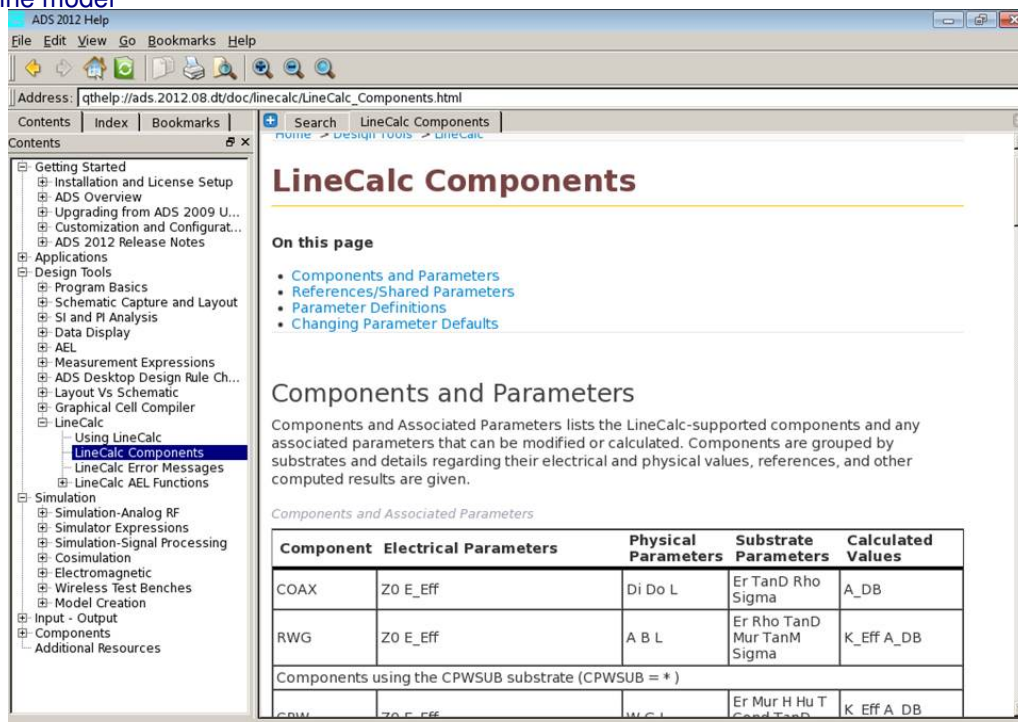
#### Applications Keysight ADS201601 examples



2.28

## 2.2 Real Lines and Components

### A line model



### LineCalc Components

On this page

- [Components and Parameters](#)
- [References/Shared Parameters](#)
- [Parameter Definitions](#)
- [Changing Parameter Defaults](#)

### Components and Parameters

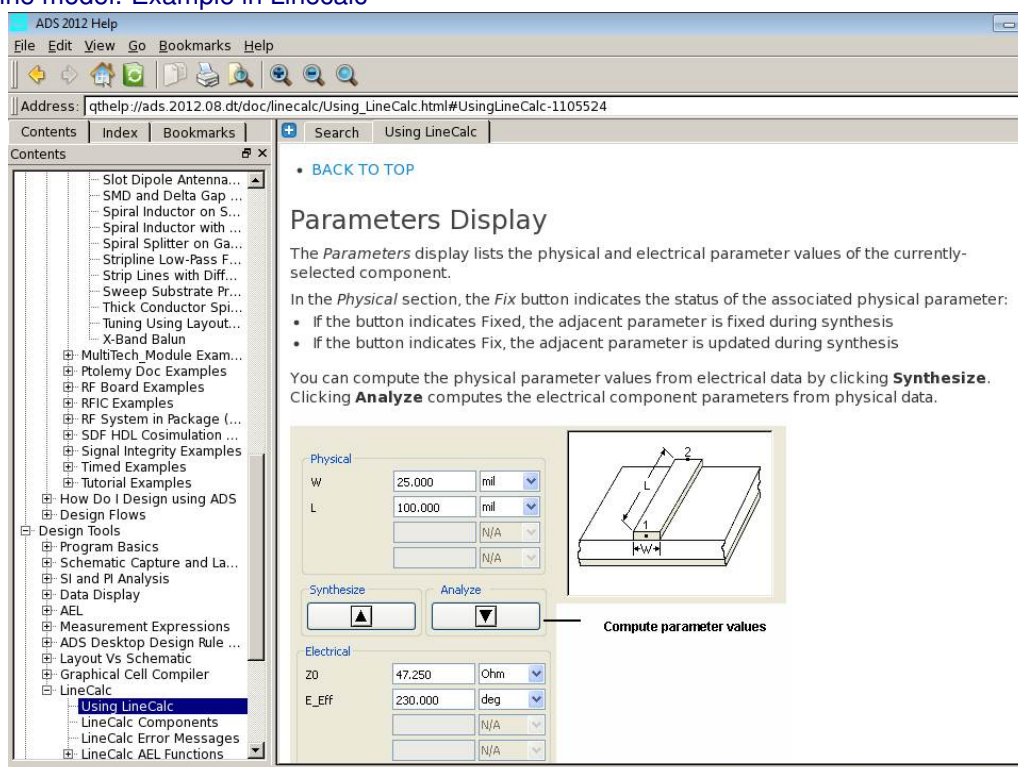
Components and Associated Parameters lists the LineCalc-supported components and any associated parameters that can be modified or calculated. Components are grouped by substrates and details regarding their electrical and physical values, references, and other computed results are given.

Components and Associated Parameters

| Component                                          | Electrical Parameters | Physical Parameters | Substrate Parameters       | Calculated Values |
|----------------------------------------------------|-----------------------|---------------------|----------------------------|-------------------|
| COAX                                               | Z0 E_Eff              | Di Do L             | Er TanD Rho Sigma          | A_DB              |
| RWG                                                | Z0 E_Eff              | A B L               | Er Rho TanD Mur TanM Sigma | K_Eff A_DB        |
| Components using the CPWSUB substrate (CPWSUB = *) |                       |                     |                            |                   |
| CPW                                                | Z0 E_Eff              | W L                 | Er Mur H Hu T              | K_Eff A_DB        |

2.29

### A line model: Example in Linecalc



### Parameters Display

The *Parameters* display lists the physical and electrical parameter values of the currently-selected component.

In the *Physical* section, the *Fix* button indicates the status of the associated physical parameter:

- If the button indicates Fixed, the adjacent parameter is fixed during synthesis
- If the button indicates Fix, the adjacent parameter is updated during synthesis

You can compute the physical parameter values from electrical data by clicking **Synthesize**. Clicking **Analyze** computes the electrical component parameters from physical data.

Compute parameter values

Physical

W 25.000 mil

L 100.000 mil

N/A

N/A

Synthesize Analyze

Electrical

Z0 47.250 Ohm

E\_Eff 230.000 deg

N/A

N/A

2.30

### Example: Coaxial Cable

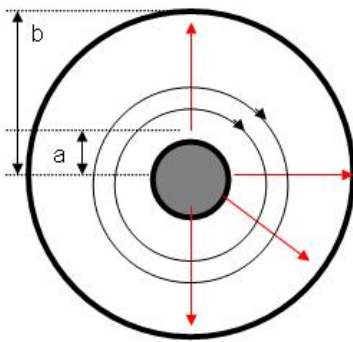
#### Coaxial example



2.31

### Example: Coaxial Cable

#### Coaxial example

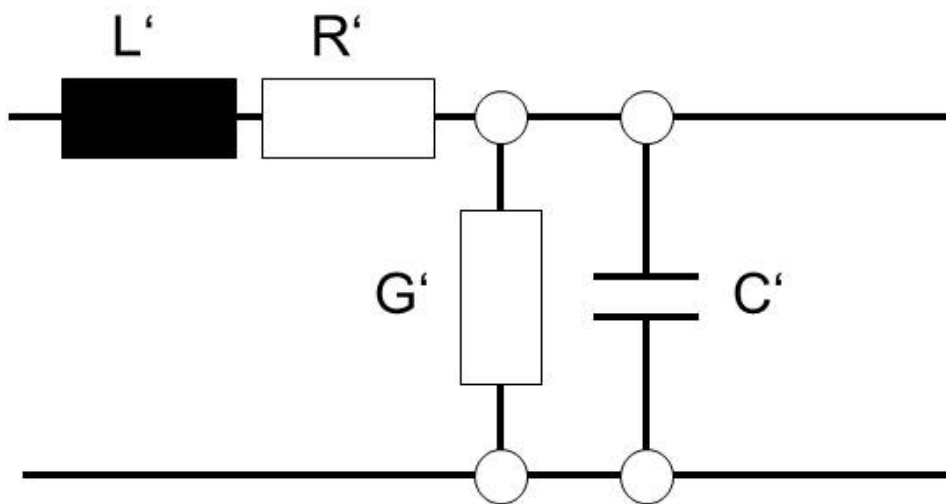


$$C' = \frac{2\pi\epsilon_0\epsilon_r}{\ln(b/a)}$$

$$L' = \frac{\mu_0\mu_r}{2\pi} \ln(b/a)$$

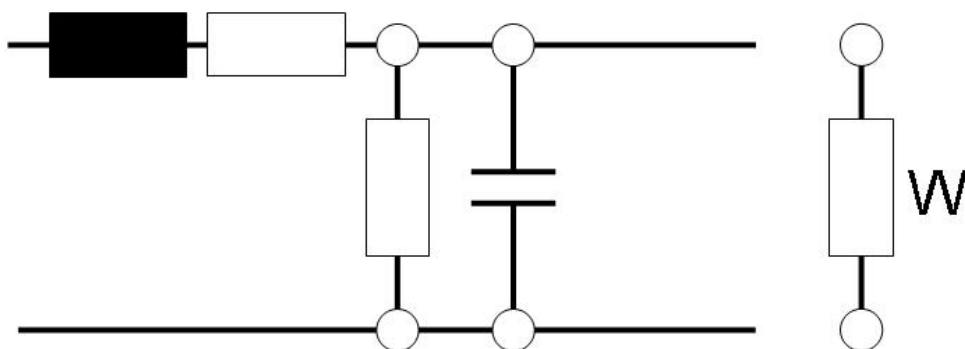
2.32

## Line models: Theory



2.33

## Line models: Theory



2.34

## A line model: Explanation 1

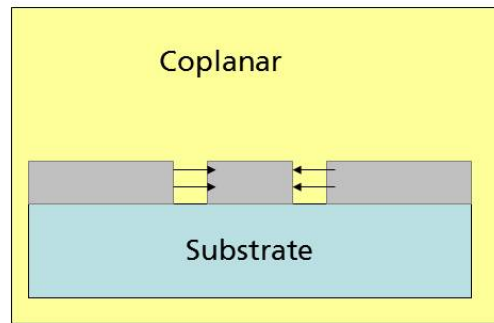
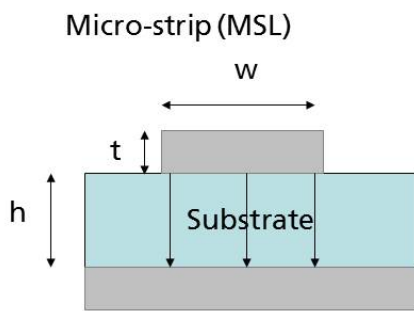
**Coaxial cable: Closed Model with the geometrical parameters  $a$  and  $b$  (inner and outer radius)**

**Effective Impedance:  $Z_0$  of the whole structure. What do you see when you enter the line ?**

**Effective Epsilon:  $\epsilon_{eff}$  over the average of the materials, weighted by the field distribution. Be careful linecalc uses this term differently !**

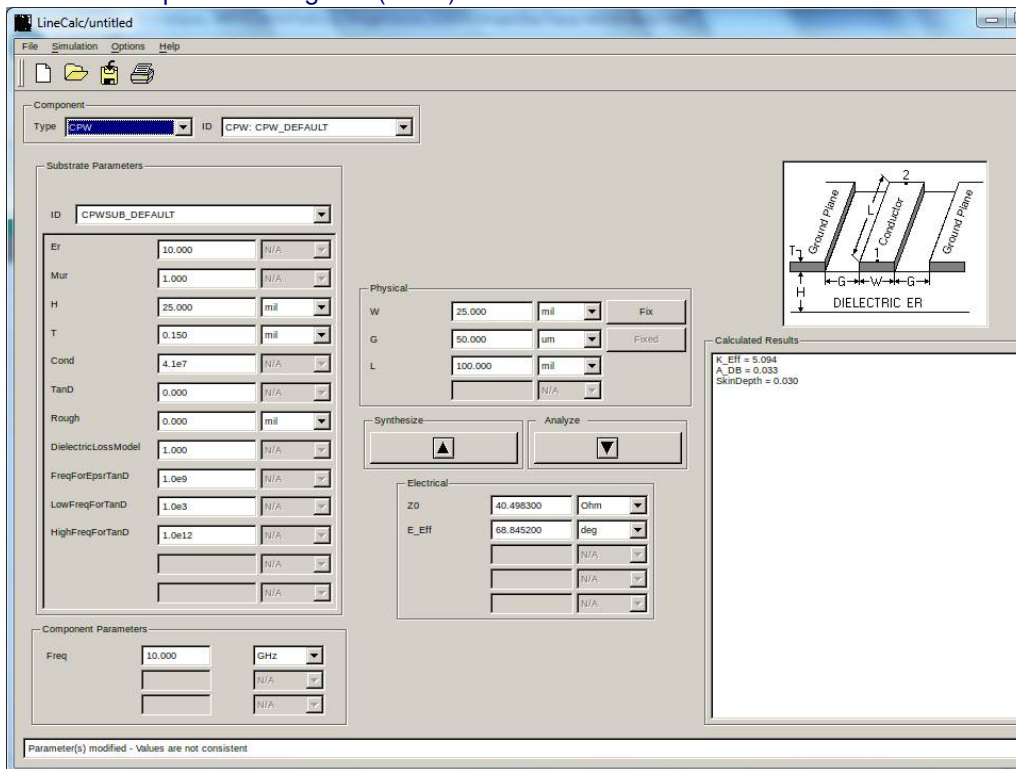
2.35

## Line models



2.36

### A line model: Coplanar waveguide (CPW)



2.37

### A line model: Explanation 2

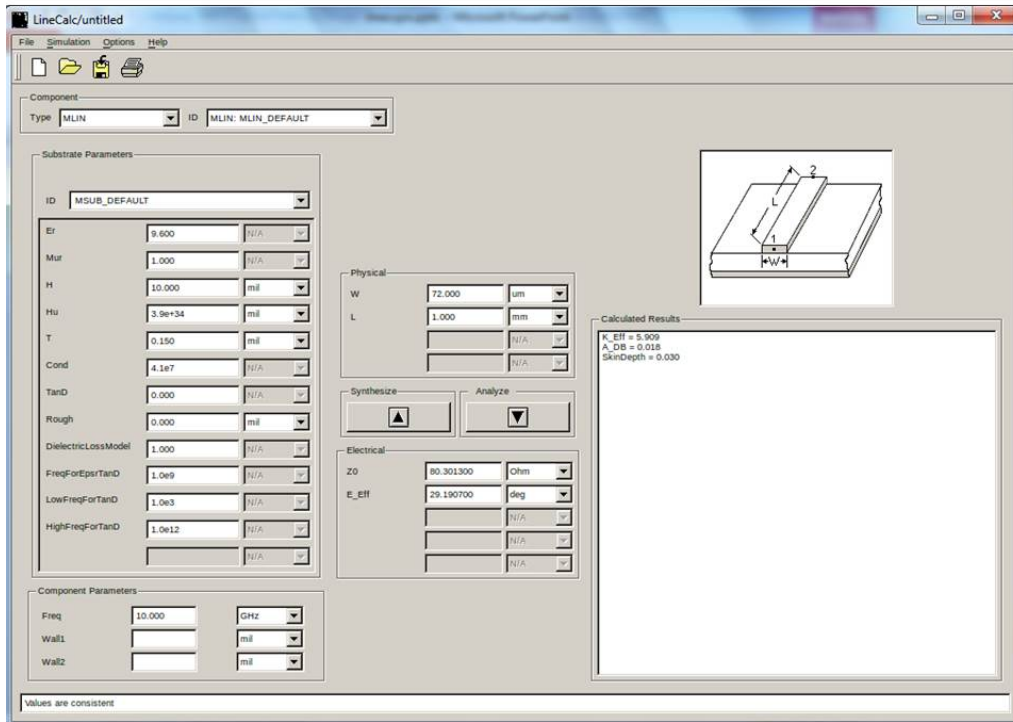
#### Coplanar model: CPW: coplanar waveguide Approximation

- GaAs or SiC semiconductor (it is up to you how many assumption you make for the roughness and the loss functions.)
- Dielectric material with little conductivity/high isolation !
- Substrate thickness: 75  $\mu\text{m}$ . Check for the name in Linecalc !
- Ground-to-ground spacing: 50  $\mu\text{m}$ . This is the distance between the two grounds, not to the hot wire.
- Gold layer: T= 5  $\mu\text{m}$ , check for the linecalc name.
- Dimension W (width of the inner conductor) to get Z0= 70 Ohms.

2.38

## A line model: Explanation 3

### Microstrip transmission line model (MSL): Model MLIN



2.39

## A line model: Explanation 3a

- SiC semiconductor epsr= 9.8
- Substrate thickness:  $H = 75 \mu\text{m}$ .
- thickness gold layer:  $T = 5 \mu\text{m}$ .

2.40

## 2.3 Task 2 (Block 1): LineCalc

### Task 2 (Block 1): LineCalc (Sum: 25 points)

1. Open workspace, Start the simulator **LineCalc**.
2. Is the impedance of a line length-dependent ? a.) In theory, b.) practically for real devices, if done correctly (two sentences each) (5 points)
3. Is the impedance of the transmission line frequency-dependent ? In case, if we specify the frequency, which parameter to design a transmission line is also strongly frequency dependent ?(5 points): First Answer yes/no. Second answer: max. three sentences !!!
4. Selection of models: Choose the right model for a coaxial line on the topic cable with ref. impedance  $Z_0 = 50 \text{ Ohm}$  impedance. Use 3 GHz for the frequency.
5. To dimension a cable, given  $D_i = 1.5 \text{ mm}$  (inner diameter), Find  $D_o$  (outer diameter). Use epsr= 3.2 for polyethylene between the conductors. (5 points) Answer: just put the number of the outer diameter. Remember radius vs. diameter.

2.41

### Task 2 (Block 1) continued: LineCalc

6. Perform a simulation of a coplanar line for epsr= 12 (GaAs) with the target for a 30 Ohm impedance. Dimension the ground-to-ground spacing (i.e. the distance of the ground-to-ground lines), assume an inner hot line width:  $10 \mu\text{m}$ . Hint: Remember was the ground-to-ground spacing is ! (5 points). Answer: give the right number of the ground-to-ground spacing.
7. Selection of transmission line: Choose again a transmission line model, this time a microstrip model (Take MLIN).
8. Dimension a  $Z_0 = 70 \text{ Ohm}$  and  $30 \text{ Ohm}$  line on SiC for a substrate thickness of  $75 \mu\text{m}$ .(5 points)

2.42

## Task 2 (Block 1) Structure Format of solution (Sum: 25 points)

**PLEASE: Use number of the block and subtasks in your answers !**

Write down the Block and the Number of the exercise, such as: Block 2, 4.)

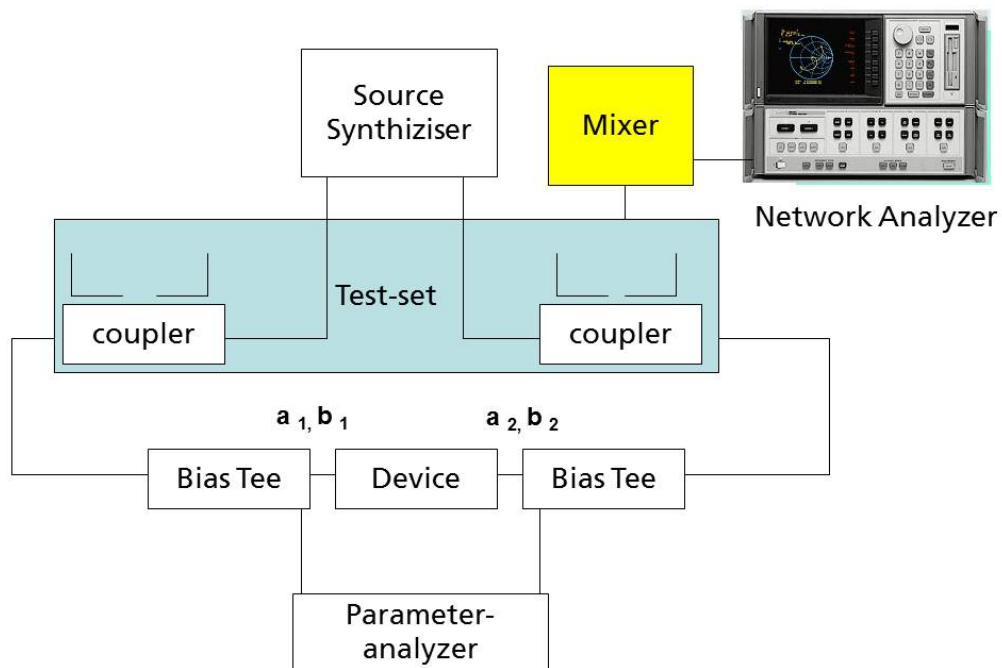
1. Number 2.) Write two sentences only (5 points)
2. Number 3.) Answer yes/no:, max. three sentences (5 points)
3. Number 5.) Put the number of the outer diameter. Remember radius vs. diameter (5 points).
4. Number 6.) Answer: give the right numbers of the ground-to-ground spacings (5 points).
5. Number 8.) Give the values for the width of the hot line for the two impedances (5 points)

**PLEASE:**

Stick to these requests, otherwise points will be reduced.

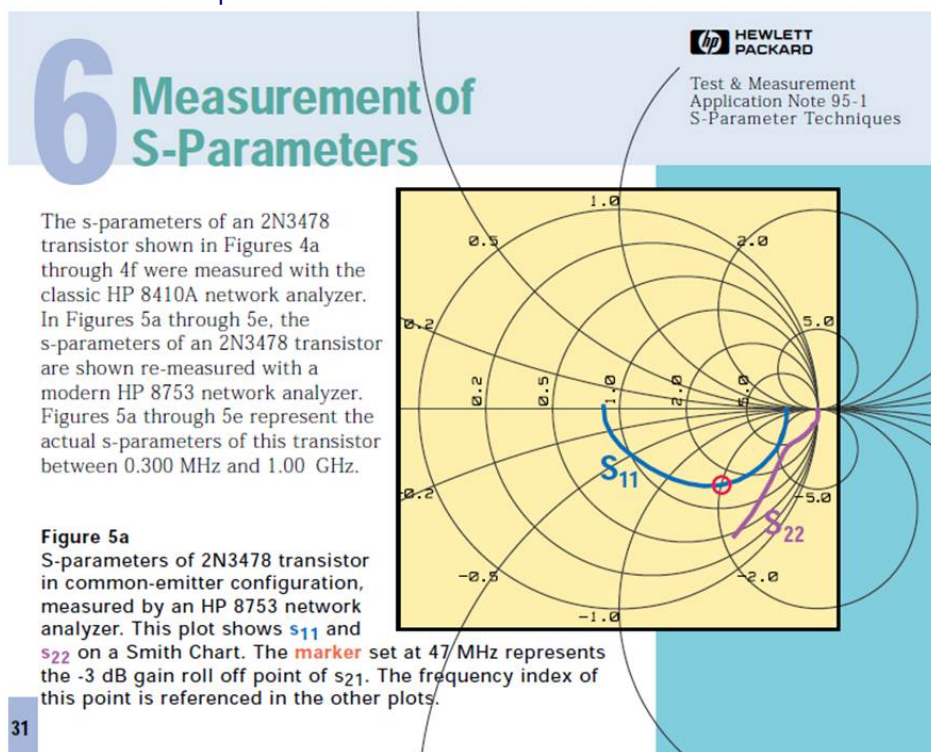
## 2.4 Microstrip and Stubs: Methods of Moments: Task 3 (Block 1)

### S-parameters: Concept



2.44

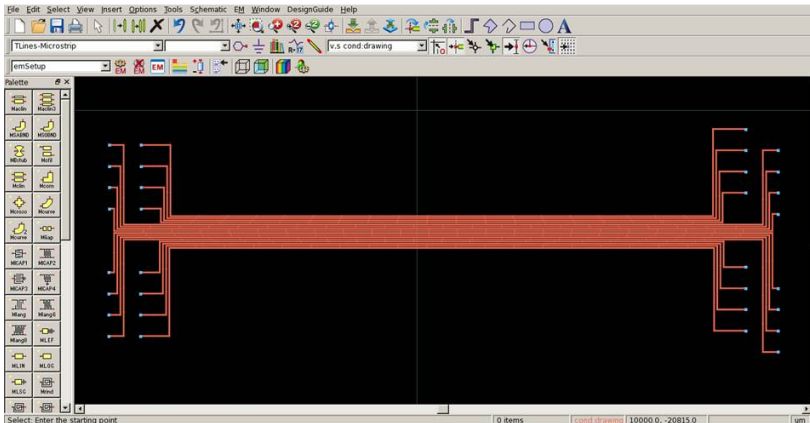
### S-parameters: Concept 2



2.45

## S-parameters: Concept 3

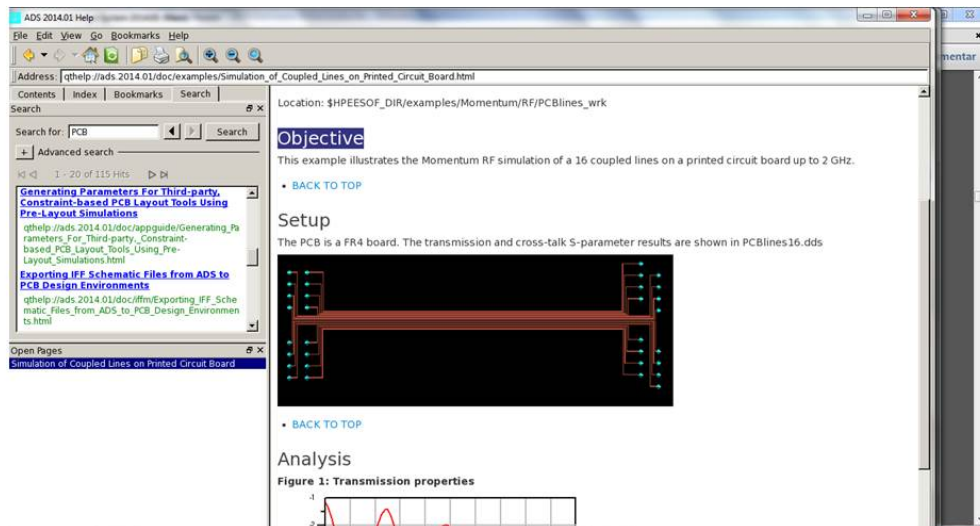
A closer look for more complex combined structures. What is different? Coupling ! Sij for i not equal j !!!



2.46

## Coupled Microstrip Lines: Task 3 (Block 1)

### Momentum Simulation

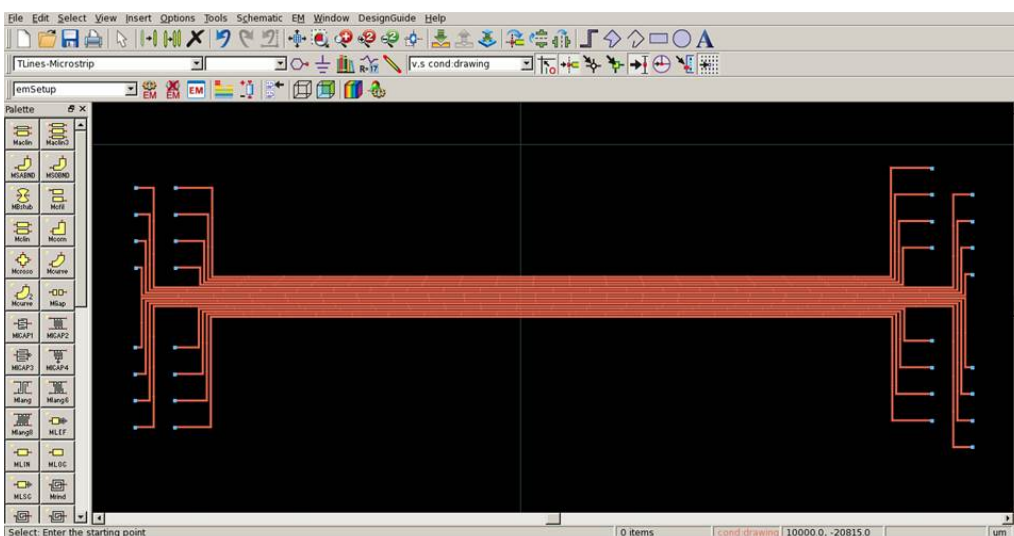


/examples/Momentum/PCLines

2.47

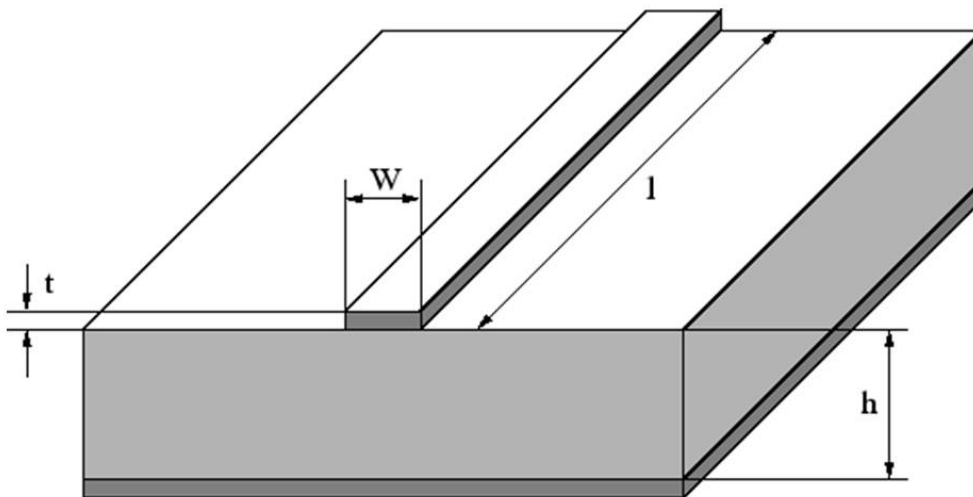
## Coupled Microstrip Lines on a Printed Circuit Board (PC)

Microstrip Lines **Couple** between the 16 ports...



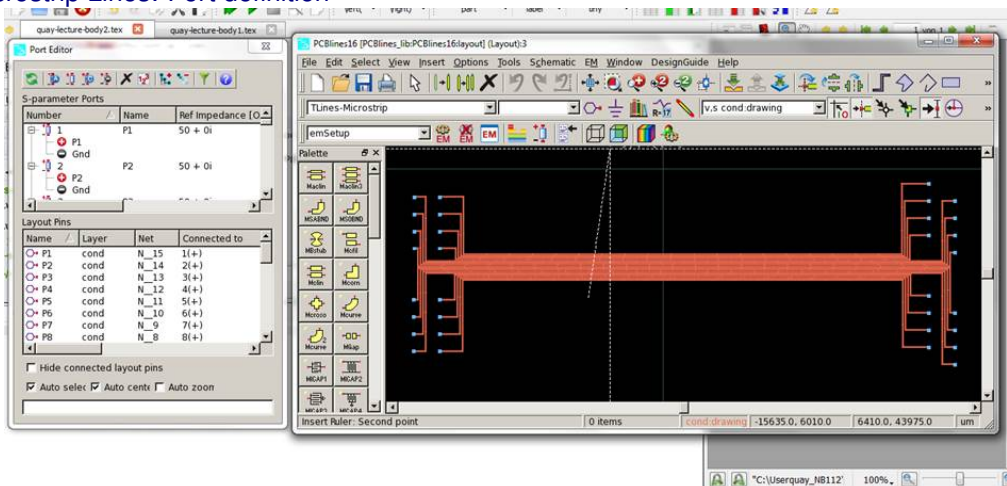
2.48

## Microstrip Lines



2.49

## Microstrip Lines: Port definition



2.50

## Task 3 (Block 1) Momentum (Sum: 25 points)

- Example: PCLines in Examples Momentum. **Single Transmission lines:**
- Give a plot for the structure with the port numbers attached to the geometry (2.5 points)
- What is the difference of Momentum over Linecalc? Give a structure example, that cannot be included in linecalc. (Write 3 meaningful sentences max. not more) (2.5 points)
- What does a general transmission line do with the impedance and the phase of the wave? (Write three sentences max. (2.5 points)
- Plot the magnitude of the transmission  $S(3,19)$  in dB 0 to 1.6 GHz. What is the reason for the resonating nature of the S-parameters on lines? Where do we have minima and maxima. Hint: Think in terms of wavelength vs. linewidth (1 Plot plus 4 sentences max.) (5 points)
- Impact of Crosstalk:**
- Plot the cross talk of  $S(1,3)$  and  $S(1,10)$  in dB with a scale -110 dB for the same frequency range to 2 GHz. Describe why there are differences between  $S(1,4)$  and  $S(1,10)$ ! (Plot and 3 sentences) (5 points)

2.51

## Task 3 (Block 1) Momentum (Sum of points: 25 points)

- An RF-power of 10 W is sent through the first transmission line starting with port 1. What is maximum power transferred to line number starting with port 10 (5 point)? (Calculation at max. three sentences)

9. How can the crosstalk technically be a.) useful and b.) where can it be harmful (one sentence each, i.e. 2 sentences) (2.5 points)

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2.52

**Format and points Task 3 (Block 1) Momentum (Sum of points: 25 points)**

1. Number 2.) (Plot, not more) (2.5 points)
2. Number 3.) (Write 3 meaningful sentences max. not more) (2.5 points)
3. Number 4.) (Write 3 meaningful sentences max. not more) (2.5 points)
4. Number 5.) (Write three sentences max. (2.5 points)
5. Number 7.) Why are they different and what exactly is different (Plot and 3 sentences ?, plot for a scale from -110 dB to 0 dB with 5 dB spacings ! (5 points)
6. Number 8.) (Calculation and 3 sentences) (5 points)
7. Number 9.) (one sentence each, i.e. 2 sentences) (2.5 points each, 5 points)

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2.53